eBOT: Depth-Aware Bandwidth-Optimized Aggregation Trees for Heterogeneous Distributed Training

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***Abstract*—Distributed machine learning is increasingly constrained by communication bottlenecks, leading to inefficient resource utilization and prolonged training times. Existing gradient aggregation approaches often rely on fixed hierarchical topologies or assume homogeneous network conditions, limiting their effectiveness in bandwidth-heterogeneous environments. This paper presents eBOT, a depth-aware, bandwidth-optimized aggregation framework that constructs multi-level aggregation trees by jointly considering bandwidth utilization and tree depth. eBOT also integrates extended Berkeley Packet Filter (eBPF) and eXpress Data Path (XDP) to support in-kernel packet processing during gradient aggregation. Simulation results show that, for 200 nodes with a bandwidth heterogeneity parameter of $\sigma = 0.10$, eBOT reduces aggregation time by 59.6% compared with ALEPH and 39.6% compared with Ring All-Reduce. These results demonstrate the benefits of adaptive aggregation topologies for distributed training under heterogeneous bandwidth conditions.**

***Index Terms*—Distributed Training, eBPF, Hierarchical Aggregation, Bandwidth Optimization**

APPENDIX

Equal-Sharing Assumption. For each node i, the available bandwidth B_i is equally shared among its $\deg(i)$ incident tree links and its local gradient flow, i.e., among $\deg(i)+1$ communication operations. Therefore, the effective bandwidth allocated to each communication operation is defined as

$$b_i = \frac{B_i}{\deg(i)+1}.$$

Proposition 1. *The additional forwarding delay of gradient fragments across aggregation levels is negligible when the number of gradient fragments is much larger than the maximum tree depth.*

Proof. Let F denote the number of gradient fragments, D the maximum tree depth, and t_{frag} the transmission time of one fragment. Under pipelined fragment transmission, the baseline communication time for transmitting all F fragments is

$$T_{\text{comm}} = F t_{\text{frag}}.$$

The additional forwarding delay introduced by traversing the aggregation levels is at most

$$T_{\text{fwd}} \le (D-1)t_{\text{frag}} = \frac{D-1}{F} T_{\text{comm}}.$$

Therefore, when $F \gg D$,

$$\frac{T_{\text{fwd}}}{T_{\text{comm}}} \le \frac{D-1}{F} \to 0.$$

Thus, the additional forwarding delay is negligible relative to the baseline fragment transmission time and can be omitted from the communication-time model. The computation overhead associated with the aggregation levels is modeled separately as

$$T_{\text{depth}} = \tau D.$$

□

Theorem 1. *Let b_i denote the normalized bandwidth of node i, defined as*

$$b_i = \frac{B_i}{\deg(i)+1},$$

and let D denote the maximum depth of the aggregation tree. Under the equal-sharing assumption, pipelined fragment transmission, and a per-level computation latency of τ, a lower bound on the total aggregation time is

$$T \ge \frac{G}{\min_{i=1}^{N} b_i} + \tau D,$$

where G denotes the gradient size.

Proof. Under the equal-sharing assumption, the effective bandwidth available to each communication operation at node i is

$$b_i = \frac{B_i}{\deg(i)+1}.$$

Since a gradient of size G must be transmitted through a communication operation with effective bandwidth b_i, the corresponding transmission time is lower bounded by

$$T_i^{\text{comm}} \ge \frac{G}{b_i}, \qquad \forall i \in \{1,\dots,N\}.$$

Therefore, the communication phase is lower bounded by its bottleneck operation:

$$T_{\text{comm}} \geq \max_{i=1}^{N} \frac{G}{b_i} = \frac{G}{\min_{i=1}^{N} b_i}.$$

By the preceding proposition, the additional forwarding delay caused by fragment traversal across aggregation levels is negligible when $F \gg D$. The computation overhead associated with the aggregation levels is modeled as

$$T_{\text{depth}} = \tau D.$$

Assuming that the communication time and the depth-dependent computation overhead contribute additively to the total aggregation time,

$$T = T_{\text{comm}} + T_{\text{depth}}.$$

Consequently,

$$T \geq \frac{G}{\min_{i=1}^{N} b_i} + \tau D.$$

□